Where to stop the magnetohydrostatic solve on an optical-depth grid

An experiment on the `logtau_cut` parameter of `mhs-poisson-ltau` in HSEMHS, tested against a Bifrost simulation. 26 October 2026.

Abstract

The mode `mhs-poisson-ltau` computes gas pressure and geometric heights on optical-depth surfaces such that pressure, gravity and the Lorentz force balance. It solves only the levels below a cut, $\log_{10}\tau_{500} \geq -4$ by default, and keeps the layers above it in column hydrostatic equilibrium. The value -4 had been chosen on physical grounds but never tested. We ran the solver on a full 504×504 Bifrost-derived atmosphere with the cut at -4, -5, -6 and with no cut, and compared each result with the simulation's own pressures and heights. The cut at -4 is the only one that converges; it reduces the error of the $\tau_{500} = 1$ surface from 85 km (column hydrostatic start) to 27 km and reproduces the Wilson depression of magnetic elements (-78 km against -74 km in the simulation). A cut at -5 is slightly worse and five times slower. Cuts at -6 and above put layers with plasma $\beta \ll 1$ into the solve; the solver then tries to balance a Lorentz force of 10–1000 times the weight of the gas with the gas pressure, and the $\tau = 1$ surface is displaced by 1–20 Mm. We recommend keeping `logtau_cut` $= -4$ and, more generally, placing the cut where the median β is still $\gtrsim 10$.

1. The problem

A spectral-line inversion gives the temperature T and the magnetic field $\mathbf{B}$ in each pixel as functions of the continuum optical depth τ_{500}, but not the gas pressure P or the geometric height z of each optical-depth surface. HSEMHS offers two ways to supply them on the τ grid:

- `column-hse-ltau`: every pixel column is put into hydrostatic equilibrium on its own, $\mathrm{d}\ln P/\mathrm{d}\ln\tau = g\tau\rho/(\chi P)$ and $\mathrm{d}z/\mathrm{d}\ln\tau = -\tau/\chi$, with χ the 500 nm extinction coefficient. Columns are not in horizontal balance with each other, and $\mathbf{J}\times\mathbf{B}$ is ignored.
- `mhs-poisson-ltau`: starting from the column solution, a Gauss–Newton (Levenberg–Marquardt) solve looks for $\ln P$ on the τ levels and the heights of the τ surfaces such that

$$-\nabla P - \rho g\,\hat{\mathbf{z}} + \mathbf{J}\times\mathbf{B} = 0, \qquad \mathbf{J} = \nabla\times\mathbf{B}/\mu_0, \tag{1}$$

  holds as well as possible in all three directions (weighted least squares), with T and $\mathbf{B}$ held fixed.

The second mode does not solve all levels. Levels above a *cut* $\log_{10}\tau <$ `logtau_cut` are kept at the column hydrostatic solution (their pressures and their distances to the cut level are fixed); only $\ln P$ below the cut and one height per column are unknowns. The default `logtau_cut` $= -4$ was set when the solver was written, by analogy with the height-grid mode, where the solve is limited to $z \leq 1$ Mm because solving into the upper atmosphere with a fixed field inflated the pressure by 1–3 orders of magnitude. The specific value -4 had not been tested.

Why a cut is needed at all. The ratio of gas to magnetic pressure, $\beta = 2\mu_0 P/B^2$, falls steeply with height. Where $\beta \ll 1$ the Lorentz force of any field that is not force-free is much larger than the weight of the gas, and the gas pressure cannot balance it. The field from an inversion (or from a dynamic simulation) is never force-free. If such a layer is inside the solved region, the least-squares solve will still try to reduce that imbalance by changing the pressure. On a τ grid this has a second effect: the height spacing of the levels, $\Delta z = -\int \tau/\chi\,\mathrm{d}\ln\tau$, depends on the pressure through $\chi(T,P)$, and every height below a level is the sum of the spacings above it. A pressure change high up therefore moves the whole column underneath, including the $\tau = 1$ surface.

Question. Is -4 a good cut? What happens at -5, -6, and with no cut at all, and what physical quantity should decide where the cut goes?

2. Data and reference truth

Input. `IO/input-mhs-ltau.nc`: T, $\mathbf{B}$ on 95 levels $\log_{10}\tau_{500} = -8.0, -7.9, \ldots, 1.4$ in 504×504 columns. The file carries no x, y axes; they were taken from the Bifrost cube it was made from (uniform spacing 47.62 km, periodic). Gravity $g = 274\,\mathrm{m\,s^{-2}}$; EOS and opacity from `cache/eos_table_fullz.h5`.

Truth. The input was derived from the Bifrost enhanced-network simulation `en024048_hion`, snapshot 385, which also holds the true P, ρ and z. We computed τ_{500} for every Bifrost column from its own T and P with the same opacity table (top-first trapezoid, as the code does), and interpolated z, $\ln P$, $\ln\rho$, $\ln T$ linearly in $\log\tau$ onto the 95 input levels. As a check, the input temperature agrees with the interpolated Bifrost temperature to 0.1–1 % (median) between $\log\tau = -4.5$ and -0.5, and to 1–9 % higher up. Individual points can differ by much more (up to a factor $\sim$2), so the input's τ scale was built with a somewhat different opacity or interpolation. *The truth is therefore reliable for statistics over many points, not point by point.* All heights are compared after removing the mean of $z(\tau = 1)$ (the code's own gauge).

3. Experiments

Five runs of the production code (main at `5e901b5`), identical except for the cut:

run	mode	`logtau_cut`	solved $\ln P$ levels per column
column HSE	`column-hse-ltau`	–	none (reference start)
cut −4	`mhs-poisson-ltau`	−4.0 (default)	54
cut −5	`mhs-poisson-ltau`	−5.0	64
cut −6	`mhs-poisson-ltau`	−6.0	74
no cut	`mhs-poisson-ltau`	−7.9 (top level, the highest allowed)	93

All other settings are the defaults: alignment of the starting columns on total pressure $P + B^2/2\mu_0$, at most 100 steps, stop when the relative decrease of the objective Φ is below 10^{-4} on three consecutive steps, height step cap 300 km. Each run used one hyades node (3–4 MPI ranks × 32–48 threads).

4. What we measured

1. **Convergence:** number of steps, stop reason, Φ/Φ_0, the Gauss–Newton decrement (relative decrease a further full step would give; small means stationary), wall time.
2. **Accuracy against Bifrost:** rms and median of $\ln(P/P_{\mathrm{Bifrost}})$ and rms of $z - z_{\mathrm{Bifrost}}$ on each level, for all columns and separately for strong-field ($|\mathbf{B}| > 500$ G at $\tau = 1$; 1.2 % of columns) and quiet (< 100 G; 91.5 %) columns.
3. **The $\tau = 1$ surface:** map of $z(\tau = 1)$, its rms error and correlation with Bifrost, and the *Wilson depression*, the mean $z(\tau = 1)$ of strong-field minus quiet columns. Magnetic elements are evacuated, so their $\tau = 1$ surface lies deeper; in this snapshot by 74 km.
4. **Pressure inflation:** $\ln(P/P_{\text{column HSE}})$ on the solved levels, i.e. how far the solve moves the pressure from the hydrostatic start.
5. **Force balance, the same for every state:** independent of the solver's own residuals and weights, we evaluate on every point the net force over the weight of the gas,

$$\frac{\mathbf{F}_h}{\rho g} = H\Big(-\nabla_h \ln P\big|_z + \frac{\mathbf{L}_h}{P}\Big), \qquad \frac{F_z}{\rho g} = H\Big(-\partial_z \ln P - \frac{\rho g}{P} + \frac{L_z}{P}\Big), \qquad H = \frac{P}{\rho g}, \tag{2}$$

   with $\mathbf{L} = \mathbf{J} \times \mathbf{B}$, horizontal derivatives at fixed height and centred differences. Applied to the Bifrost truth this gives the imbalance a real, dynamic atmosphere has (Bifrost has flows, so it is not static).
6. **Plasma β and $|\mathbf{L}|/\rho g$** per level (from Bifrost), the scale of what the field can do compared with gravity.

5. Results

5.1 Summary

run	steps	stop	Φ/Φ_0	wall	$z(\tau{=}1)$ err (rms)	corr.	Wilson depr.
column HSE	–	–	–	1.3 min	85 km	0.61	−106 km
cut −4	19	converged	1.8×10^{-2}	16 min	**27 km**	**0.69**	**−78 km**
cut −5	100	max_steps	2.2×10^{-3}	89 min	40 km	0.62	−104 km
cut −6	100	max_steps	1.1×10^{-1}	64 min	1 137 km	0.09	−180 km
no cut	100	max_steps	1.8×10^{-1}	81 min	20 184 km	0.06	−16 Mm
Bifrost					0	1	−74 km

Φ_0 itself grows by fifteen orders of magnitude as the cut moves up (7.4×10^{10}, 1.2×10^{13}, 3.1×10^{17}, 2.9×10^{25} for $-4, -5, -6$, none): the objective is increasingly dominated by the force imbalance of the upper layers. Ratios Φ/Φ_0 are therefore not comparable between runs.

5.2 Where the field takes over

	Bifrost			median $\lvert\mathbf{F}_h\rvert/\rho g$					
$\log\tau$	median β	frac. $\beta<1$	$\lvert\mathbf{L}\rvert/\rho g$	Bifrost	col. HSE	cut -4	cut -5	cut -6	no cut
-6.0	0.014	0.98	14.6	11.1	8.6	8.0	8.0	8.4	21.5
-5.5	0.15	0.73	2.9	2.1	0.64	0.35	0.37	1.8	5.6
-5.0	3.0	0.35	0.18	0.21	0.36	0.14	0.14	1.3	4.4
-4.5	10.7	0.13	0.080	0.10	0.29	0.088	0.068	1.2	4.2
-4.0	21	0.05	0.050	0.073	0.27	0.049	0.056	1.1	4.1
-3.0	162	0.017	0.013	0.054	0.24	0.021	0.047	1.1	4.1
-2.0	4 200	0.005	0.0006	0.055	0.23	0.018	0.045	1.1	4.1
-1.0	17 900	0.002	0.0001	0.071	0.21	0.027	0.054	1.1	4.1
0.0	23 200	0.001	0.0001	0.098	0.15	0.045	0.065	1.0	4.1

The median β crosses 1 at $\log\tau \approx -5.5$. At -4 the gas dominates almost everywhere ($\beta \approx 21$, 5 % of points below 1, Lorentz force 5 % of the weight). At -5 a third of the points already have $\beta < 1$; above -6 practically all do and the Lorentz force is 15–350 times the weight.

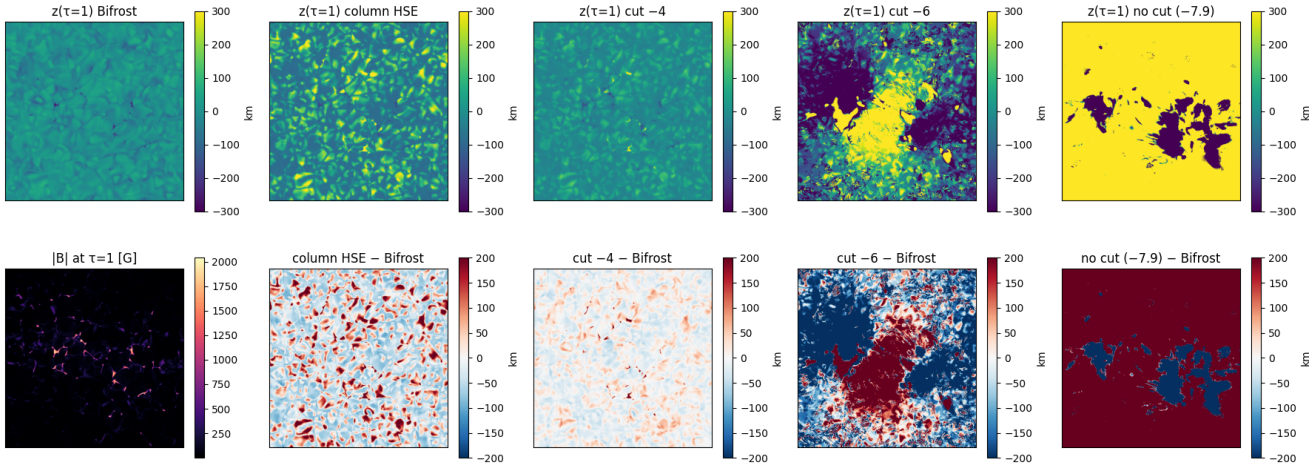


Figure 1: Top: height of the $\tau_{500} = 1$ surface (each map's mean removed) in Bifrost, column HSE, and the cut -4, -6 and no-cut solutions; bottom left: $|\mathbf{B}|$ at $\tau = 1$; bottom: difference from Bifrost. Colour ranges ± 300 km (top) and ± 200 km (bottom) saturate for cut -6 and no cut.

5.3 The $\tau = 1$ surface (Fig. 1)

Column HSE gives the granulation pattern with exaggerated contrast: every column is hydrostatic on its own and aligned only through one total-pressure level, and many structures sit 100–200 km too high or too low. The cut -4 solve removes most of this: the remaining error is small and unstructured, the rms error drops threefold and the Wilson depression of the magnetic elements comes out at -78 km against -74 km in Bifrost (column HSE: -106 km). With the cut at -6 the surface is pushed up and down by up to several Mm in patches tens of Mm across; with no cut nearly the whole surface is shifted by Mm, the mean gauge then hiding a few deep regions (down to -60 Mm at the 1st percentile).

5.4 Profiles (Fig. 2)

- **Heights** (top right): between $\log\tau = -3$ and 0 the height error is 20–30 km for cut -4, 30–40 km for cut -5, 70 km for column HSE (all rise below $\tau = 1$), ∼1 Mm for cut -6 and ∼20 Mm for no cut. Because heights are sums of the spacings above, an error made high in the column is carried unchanged to every level below: the curves are flat below the layer where the damage happens.
- **Pressure** (top left and middle): between $\log\tau = -3$ and 0 cut -4 has the smallest rms error (0.04–0.1, i.e. 4–10 %); all runs share the same large error above -4 (rms 1.5–3, a factor 5–20), because the τ-hydrostatic upper atmosphere does not describe Bifrost's dynamic chromosphere and transition region (the input temperature also deviates most there). Near and below $\tau = 1$ every run has errors of 0.3–0.6, which are equally present in column HSE: they come from the τ scale mismatch and from Bifrost's convective pressure fluctuations, not from the cut.
- **Inflation** (bottom left): the higher the cut, the more the solve raises the pressure just below it (99th percentile up to $e^{2.7} \approx 15$ times the hydrostatic value at the -6 cut).

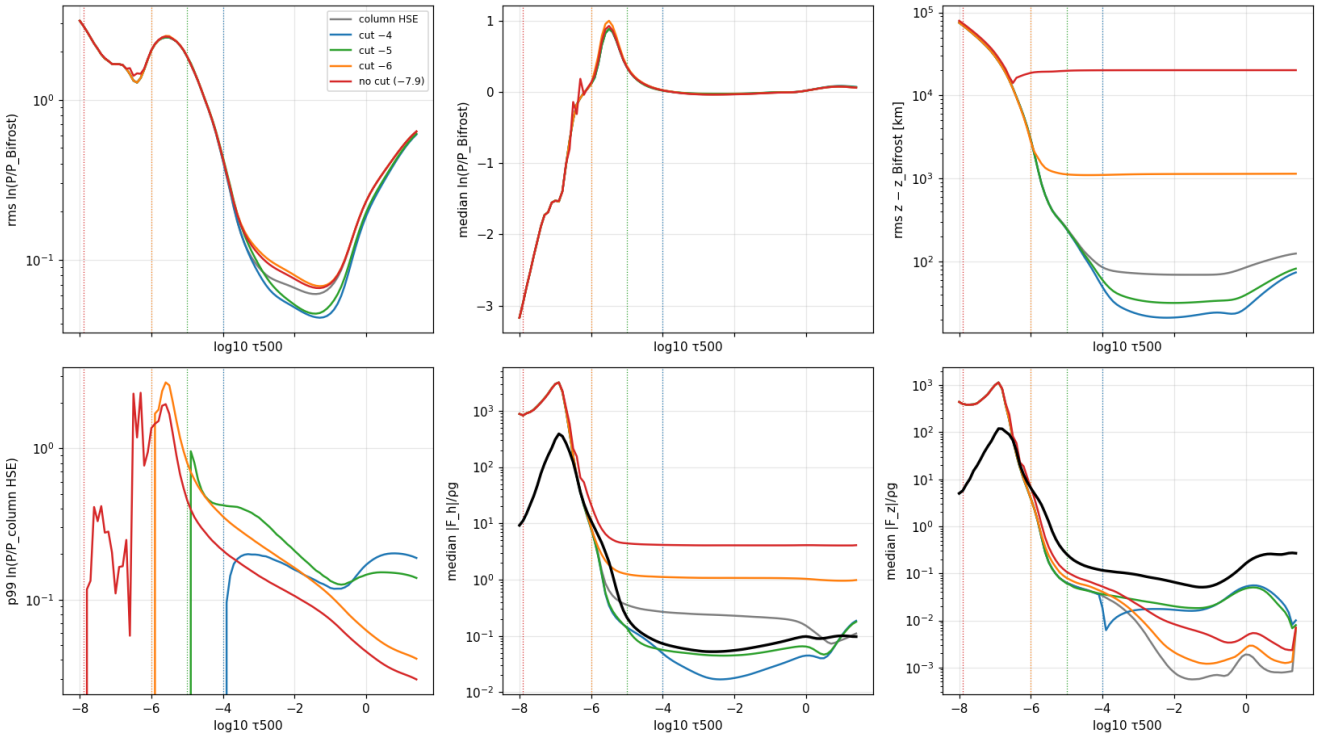


Figure 2: Per-level statistics against $\log_{10}\tau_{500}$. Top: rms and median of $\ln(P/P_{\mathrm{Bifrost}})$; rms of $z - z_{\mathrm{Bifrost}}$. Bottom: 99th percentile of $\ln(P/P_{\mathrm{column\,HSE}})$ (pressure inflation by the solve; zero above each cut by construction); median net horizontal and vertical force over ρg. Black: Bifrost. Dotted verticals: the cut of the run with the same colour.

- **Force balance** (bottom middle and right): cut -4 and -5 bring the horizontal imbalance in the photosphere from 0.2–0.3 ρg (column HSE) to 0.02–0.06 ρg, which is *below* Bifrost's own 0.05–0.1 ρg. A static model should not be expected to do better than this. Cut -6 and no cut end at 1 and 4 ρg everywhere below the cut: worse than doing nothing.

5.5 Error against β

Figure 3 shows the distribution of $\ln(P/P_{\mathrm{Bifrost}})$ against the Bifrost β for all points. At $\beta > 1$ the errors are concentrated around zero for all runs; the large errors are at $\beta < 1$, i.e. in the fixed upper part, and are inherited from the hydrostatic start. The point-wise pressure error is thus similar for all cuts; what differs is the geometry (Fig. 1), which is where a high cut does its damage.

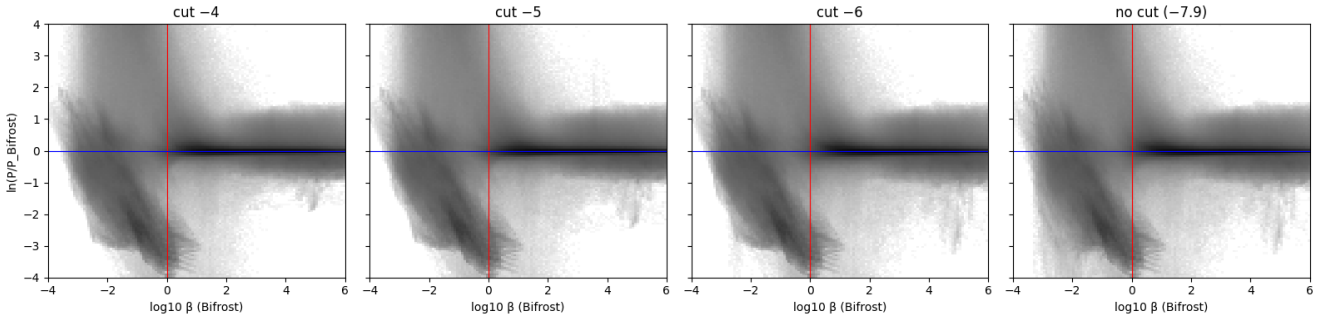


Figure 3: Two-dimensional histogram (log counts) of $\ln(P/P_{\mathrm{Bifrost}})$ against $\log_{10}\beta$ (Bifrost), all levels and columns. Red line: $\beta = 1$.

6. Interpretation

1. **Why the high cuts fail.** The input field is not force-free (it comes from a dynamic simulation; an inverted field will not be force-free either). In the layers with $\beta \ll 1$ the residual Lorentz force is 10–1000 ρg. A least-squares solve with only the pressure free cannot remove it; it reduces it partially by large pressure changes. These changes alter the opacity, hence the height spacing of the levels, and every height below moves with them. The objective is dominated by these upper residuals (15 orders of magnitude in Φ_0), so the solver accepts a much

worse photosphere in exchange for a small relative gain above. The step lengths stay tiny (line search $\alpha \approx 0.02$ for no cut) and none of these runs converge in 100 steps.

2. **Why -4 works.** Down to $\log\tau = -4$ the gas dominates ($\beta \approx 20$, Lorentz force $\approx 5\,\%$ of the weight), so the pressure can absorb the remaining imbalance with small changes, and the solved region contains the photosphere where the inversion constrains T and $\mathbf{B}$ best.
3. **Why -5 is not better.** It adds the layer where β falls through 1–10 and 35 % of points are already magnetically dominated. The solve still converges toward a reasonable state but slowly (not converged in 100 steps, 89 min), and those extra levels pull the $\tau = 1$ heights of magnetic elements too deep (Wilson depression -104 km).

7. Conclusions

- Keep `logtau_cut` $= -4$. It is the only cut tested that converges, it is the fastest of the solves, and it gives the best heights ($z(\tau{=}1)$ error 27 km, Wilson depression within 4 km of the simulation) and a photosphere in better force balance than the simulation itself.
- The cut must lie in the gas-dominated layer. A practical rule: put it where the median β of the starting state is still $\gtrsim 10$ and only a few percent of points have $\beta < 1$. For a strongly magnetised field of view (sunspots, plage) this level lies deeper and the cut should be moved down accordingly; for this enhanced-network snapshot it is $\log\tau \approx -4$.
- Solving without a cut, or with the cut in the low-β layer, is harmful: the pressure absorbs an unbalanceable Lorentz force and displaces the whole $\tau = 1$ surface by Mm.
- The pressure above the cut is not reliable in any run (off by a factor 5–20 against Bifrost): hydrostatic equilibrium in τ does not describe a dynamic chromosphere. Results above $\log\tau \approx -4$ should not be interpreted as physical pressures.

Limitations. One snapshot of one simulation (enhanced network, weak fields: 1.2 % of columns above 500 G at $\tau = 1$); cuts tested only at whole decades; the reference truth on the τ grid is only statistically consistent with the input (Sect. 2). A sunspot or plage field of view, and a β-based automatic cut, would be the natural next tests.

Reproduction. Branch `ltau-cut-study`: `HSEMHS/scripts/ltaucut/truth_on_tau.jl` (truth on the τ grid), `force_diag.jl` (force balance), `analyse.py` (statistics and figures); configurations, logs and outputs in `HSEMHS/out/ltaucut/` of the cluster worktree `MHS_ltaucut`.